

CSE451 Computer & Network Security

Assignment 1: Data Encryption Standard (DES)

Team size: up to **5** members.

It is required to implement the DES algorithm in C.

DES supplementary material:

https://en.wikipedia.org/wiki/DES_supplementary_material

Required build command:

```
gcc -O3 studentID.c -o studentID
```

The submission is one C source file. No need to use any external libraries or headers.

Required program usage:

```
studentID.EXE "e" key plaintext ciphertext
studentID.EXE "d" key ciphertext plaintext
```

The command arguments (key, plaintext, and ciphertext) are all filenames. Use fopen(), fread(), fwrite(), and fclose() to access the data.

Use ECB block cipher mode: Divide the plaintext and ciphertext file into blocks and encode each block separately with the same key. Assume the message size is divisible by 8 bytes, and no padding is needed.

Declare the main function like this:

```
int main(int argc, char **argv)
{
        return 0;
}
```

Then add your code.

Program speed is measured with the encryption and decryption of a large message. The provided message size is always a multiple of the block size, so do not implement any padding techniques.

Please mention the names and IDs of all team members in a comment at the start of your file.

Grading criteria:

Marks out of 10	Case
5	Compile error
7	Runtime error or wrong CLI
8	Wrong output
10	Correct output